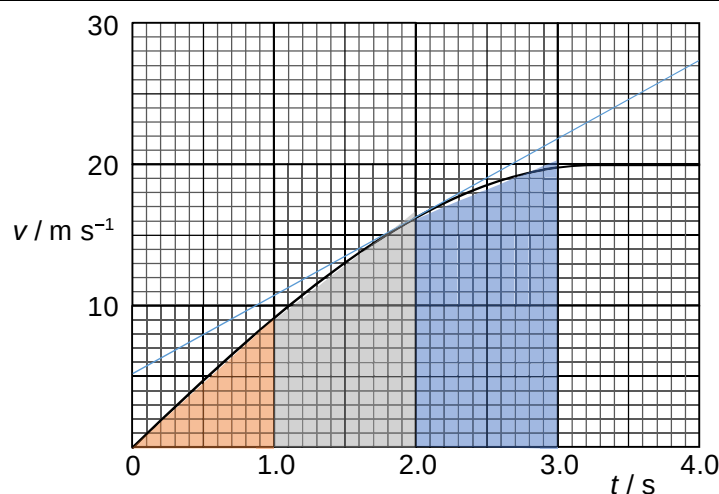


2	(a)		As the towel <u>speeds up</u>, the <u>drag force increases</u>. When the <u>drag force is equal and opposite to the weight</u> OR <u>upward drag force is equal to weight</u>, the <u>net force is zero</u> OR <u>acceleration of towel is zero</u>. and towel falls at constant speed. [if no mention of “upward” or “opposite”, max 2 marks]	B1 M1 A1
2	(b)	(i)	Show working for gradient, with correct substitution of values from graph. Correct calculation for each acceleration $a_{20 \text{ m s}^{-1}} = \frac{24 - 9}{3.4 - 0.7} = 5.56 \text{ m s}^{-2}$ accept : 5.0 – 6.1	[M1] [A1]



2	(b)	(ii)	$\text{area} = \frac{1}{2}(1.0)(9) + \frac{1}{2}(1.0)(9 + 16.5) + \frac{1}{2}(1.0)(16 + 20) + (1.0)(20)$ $= 55.25 \text{ m}$ $= 55 \text{ m}$ accept : 52 – 58 m	M1 A1
2	(c)	(i)	$s_y = u_y t + \frac{1}{2} a_y t^2$ $-29 = (8.4 \sin 40^\circ) t + \frac{1}{2} (-9.81) t^2 \quad \text{[M1]}$ $t = 3.04 \text{ s} \quad \text{or} \quad t = -1.94 \text{ s (reject)} \quad \text{[A1]}$	
	(c)	(ii)	$S_x = u_x t$ $= (8.4 \cos 40^\circ)(3.0434)$ $= 19.6 \text{ m} \quad \text{[A1]}$	